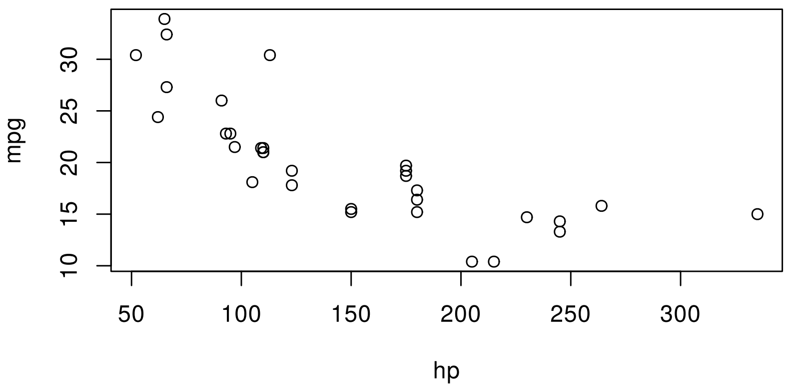
Figures and tables

A chunk that draws a plot becomes a figure, and a chunk that prints Typst markup becomes a table. *Calepin* runs the code, collects the output, and renders it back into the document in place.

Figures

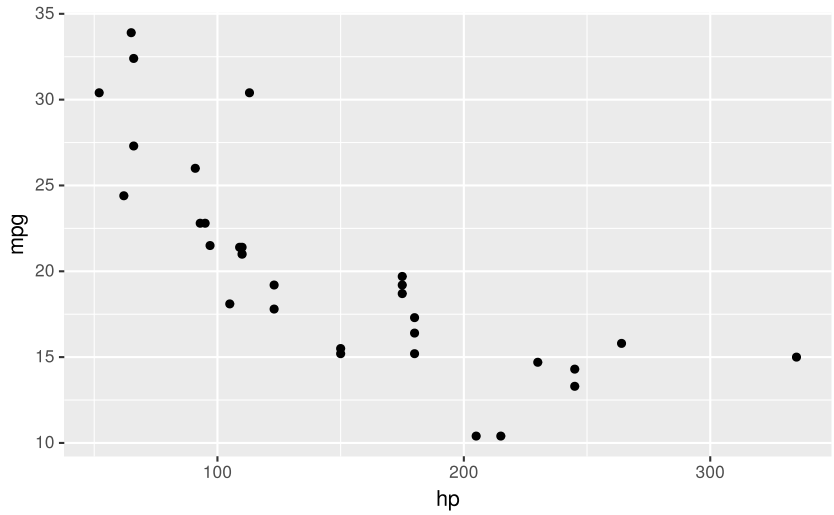
Any chunk that produces a plot is shown as a figure. Nothing extra is required:

```
plot(mpg ~ hp, data = mtcars)
```



`ggplot2` works the same way:

```
suppressPackageStartupMessages(suppressWarnings(library(ggplot2)))
ggplot(mtcars, aes(hp, mpg)) +
  geom_point()
```



Captions and labels

Add `fig-caption` to wrap the plot in a numbered figure. Add a `fig- label` so you can refer to it from the prose with `@fig-mpg` (see Cross-references).

````
#calepin.chunk(label: "fig-mpg", fig-caption: [Mileage and horsepower])[
```r
plot(mpg ~ hp, data = mtcars)
```
]
````

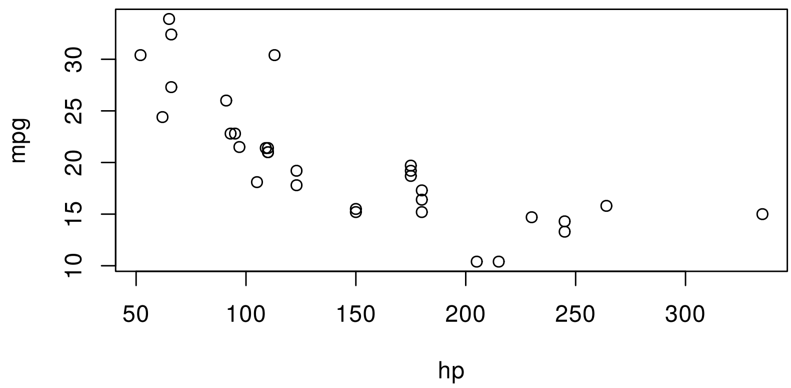
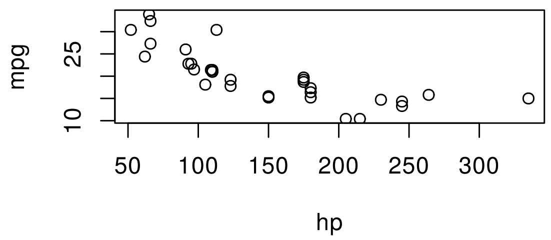


Figure 1: Mileage and horsepower

Size and alignment

Set the rendered size with `fig-width` and `fig-height`, and the placement with `fig-align`. Sizes accept a Typst length or ratio, such as `50%` or `12cm`.

````
#calepin.chunk(fig-width: 50%, fig-align: left)[
```r
plot(mpg ~ hp, data = mtcars)
```
]
````



In HTML output, `fig-responsive` (on by default) lets a figure shrink to fit a narrow screen.

Display vs. device

A figure has two sizes that do different jobs. The *device* is the canvas the engine draws on, set in inches with `fig-device-width`, `fig-device-height`, and `fig-device-aspect`. The *display* size is how large the finished image appears in the document, set with `fig-width` and `fig-height`.

Text, points, and line widths are drawn at fixed sizes on the device, and the whole canvas is then scaled to the display size. A larger device fits more inches into the same displayed width, so these elements end up looking smaller; a smaller device makes them look larger. To enlarge labels and points, shrink the device rather than the display width.

Here is the same plot on a wide device and a narrow one, both shown at the same display width. Only `fig-device-width` differs:

````
#calepin.chunk(fig-device-width: 8, fig-width: 55%)[
```r
plot(mpg ~ hp, data = mtcars, pch = 19)
```
]
````

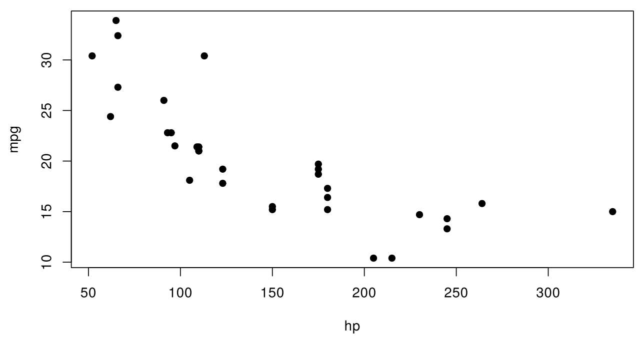


Figure 2: Wide device (8 in): smaller text and points

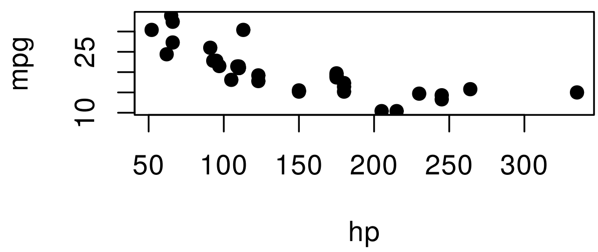


Figure 3: Narrow device (4 in): larger text and points

See Code execution for the full list of device settings.

Image format

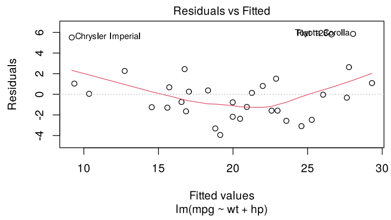
Figures are written as SVG by default. Switch to a raster format with `fig-device-format`, and set its resolution with `fig-device-dpi`:

````
#calepin.chunk(fig-device-format: "png", fig-device-dpi: 200)[
```r
plot(mpg ~ hp, data = mtcars)
```
]
````

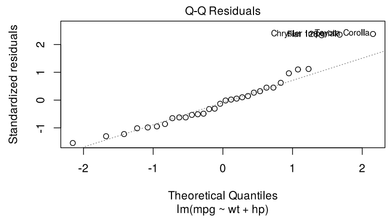
Multiple panels

A chunk that draws several plots can lay them out as one figure. Use `fig-layout-columns` and `fig-layout-rows` for the grid, and `fig-subcaptions` for per-panel captions:

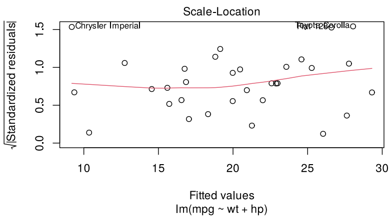
```
#calepin.chunk(
  label: "fig-diagnostics",
  fig-caption: [Regression diagnostics],
  fig-subcaptions: (
    [Residuals vs fitted],
    [Normal Q-Q],
    [Scale-location],
    [Cook's distance],
  ),
  fig-layout-columns: (1fr, 1fr),
)[```r
model <- lm(mpg ~ wt + hp, data = mtcars)
plot(model, which = 1)
plot(model, which = 2)
plot(model, which = 3)
plot(model, which = 4)
```]
```



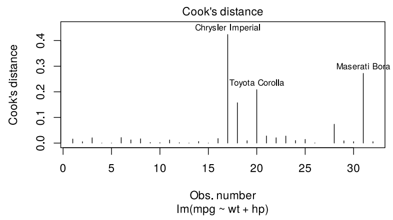

Residuals vs fitted




Normal Q-Q




Scale-location




Cook’s distance

Figure 4: Regression diagnostics

With two columns, the four plots fill a two-by-two grid. The first plot is saved under the chunk label; later plots use numbered names such as `fig-diagnostics-2.svg`.

# Tables

A table is Typst markup produced by your code. Set `results: "typst"` so *Calepin* passes that markup through to the document instead of showing it as plain text.

The R tinytable package can print a styled table as Typst:

```
#calepin.chunk(results: "typst")[
```r
library(tinytable)
tt(head(iris), caption = "First rows of iris") |>
```

```
  style_tt(i = 1:2, j = 2:3, background = "teal", color = "white")
```
]
```

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|---|---|---|---|---|
| 5.1 | 3.5 | 1.4 | 0.2 | setosa |
| 4.9 | 3.0 | 1.4 | 0.2 | setosa |
| 4.7 | 3.2 | 1.3 | 0.2 | setosa |
| 4.6 | 3.1 | 1.5 | 0.2 | setosa |
| 5.0 | 3.6 | 1.4 | 0.2 | setosa |
| 5.4 | 3.9 | 1.7 | 0.4 | setosa |

Table 1: First rows of iris

The default `tinytable` Typst output works in both PDF and HTML output.
```